BBA · Applied Mathematics for Business · 2026–2027

Unit 2 Problem Book

One-Variable Calculus · Topics 2.1 – 2.5

Problems 2.1	Functions, limits and continuity (8 problems)
Problems 2.2	Derivatives (8 problems)
Problems 2.3	Function analysis and optimization (8 problems)
Problems 2.4	Taylor approximations (7 problems)
Problems 2.5	Integration (10 problems)
Past-paper exam C	4 long problems · 90 minutes
Past-paper exam D	4 long problems · 90 minutes
Problem answers	Full worked solutions, at the end

How to use this book

1. These are full calculation problems in the style of the Practice and Team Problems. Solve every part on paper and show all steps.
2. Justify continuity, differentiability and extrema with limits and derivatives, *not* with the graph, unless the problem gives you a graph.
3. Finish every application with a sentence in context, with units.
4. Only then compare with the Problem Answers at the back. Mark each problem you got wrong and redo it tomorrow without looking.

Problems · Topic 2.1

UNIT 2. ONE-VARIABLE CALCULUS

TOPIC 2.1. FUNCTIONS, LIMITS AND CONTINUITY

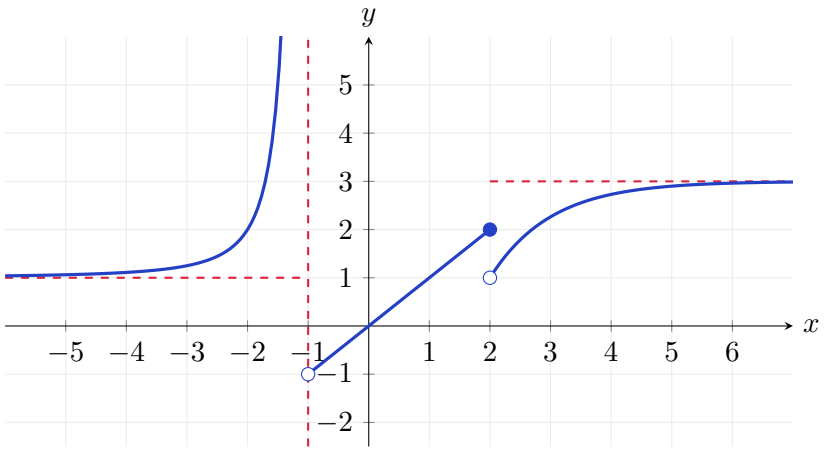
Problems · Answers at the end of the book

Problem 1 — (Definitions · equilibrium)

The supply and demand functions of a product are $q_{sup} = 3p^2$ and $q_{dem} = 375\,p^{-1}$, where p is the price. Find the equilibrium price and quantity.

Problem 2 — (Graphs of functions · limits from a graph)

The figure shows the graph of $f(x)$ and its asymptotes (dashed red lines). A filled dot means the point belongs to the graph, an empty dot means it does not.



Calculate the following limits, justifying the answers:

(i) $\lim_{x\to -1^-} f(x)$

(ii) $\lim_{x\to -1^+} f(x)$

(iii) $\lim_{x\to 2^-} \big(f(x)+1\big)$

(iv) $\lim_{x\to 2} f(x)$

(v) $\lim_{x\to -\infty} f(x)$

(vi) $\lim_{x\to +\infty} \dfrac{1}{f(x)}$

Then classify the discontinuities of f.

Problem 3 — (Definitions · limits · interpretation)

The cost, in euros, of removing $x\%$ of the plastic waste from a beach is

$$C(x) = \frac{20\,000\,x}{100 - x}.$$

a) Find the mathematical and the contextual domain of C.

b) Find the cost of removing half of the plastic.

c) What percentage of the plastic can be removed with €80 000? And with €180 000?

d) Evaluate and interpret $\lim_{x\to 100^-} C(x)$.

Problem 4 — (General functions · continuity with parameters)

Let

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x < 1 \\ ax + 3 & \text{if } 1 \le x \le 4 \\ \ln(x-3) + b & \text{if } x > 4 \end{cases}$$

a) Find the values of a and b such that $f(x)$ is continuous at $x = 1$ and at $x = 4$ (do not use a graph as a justification).

b) Take $a = 1$ and $b = 0$. Study the continuity of f and classify any discontinuity.

Problem 5 — (New functions from old · composition and domains)

Given $f(x) = \sqrt{x-2}$ and $g(x) = \dfrac{1}{x-5}$:

a) Calculate $h(x) = (f \circ g)(x)$ and $m(x) = (g \circ f)(x)$.

b) Determine the domain of $h(x)$ and of $m(x)$.

Problem 6 — (Linear and quadratic models · economic application)

A firm sells q units at the price $p(q) = 90 - 0.5q$ euros per unit. Its total cost is $C(q) = 600 + 20q + 0.5q^2$ euros.

a) Determine the contextual domain of $p(q)$.

b) Determine the income function $I(q)$ and the profit function $\pi(q)$.

c) Find the break-even quantities. For which quantities does the firm make a profit?

d) The average cost is $A(q) = \dfrac{C(q)}{q}$. Calculate $\lim\limits_{q \to 0^+} A(q)$ and interpret it.

Problem 7 — (Limits at infinity · long-run behaviour)

The number of users of a new app after t months is modelled by

$$N(t) = \frac{2000}{1 + 9e^{-0.5t}}, \qquad t \ge 0.$$

a) How many users does the app have at launch?

b) Calculate $\lim\limits_{t \to +\infty} N(t)$ and interpret it. What kind of asymptote is this?

c) A second model is $M(t) = 500\left(1 - e^{-0.2t}\right)$. Calculate $\lim\limits_{t \to +\infty} M(t)$ and $\lim\limits_{t \to +\infty} \big(N(t) - M(t)\big)$.

Problem 8 — (Floor function · discontinuous models)

A car park charges a €3 entry fee plus €2 for every *completed* hour. The price for a stay of t hours, $0 < t \le 5$, is $P(t) = 3 + 2\lfloor t \rfloor$.

a) Calculate $P(0.5)$, $P(2.5)$ and $P(3)$.

b) Find the range of P.

c) Find the points of discontinuity of P in $(0, 5)$ and classify them using one-sided limits.

d) Give one more business example where a discontinuous model describes reality better than a continuous one.

Problems · Topic 2.2

UNIT 2. ONE-VARIABLE CALCULUS

TOPIC 2.2. DERIVATIVES

Problems · Answers at the end of the book

Problem 1 — (Simple rules · quotient rule · monotonicity)

The cost, in euros, of removing x% of the plastic waste from a beach is $C(x) = \dfrac{20\,000\,x}{100 - x}$ (Topic 2.1, Problem 3).

a) Prove that $C(x)$ is increasing on all its contextual domain.

b) Calculate $C'(50)$ and $C'(90)$ and interpret both values, with units. What do they tell you about cleaning the last few percent?

Problem 2 — (Slopes · tangents · economic applications)

A firm's total cost, in euros, of producing q units is

$$C(q) = 0.01q^3 - 0.6q^2 + 30q + 400, \qquad q \geq 0.$$

a) Calculate the average rate of change of total cost when production increases from 10 to 20 units.

b) Find the marginal cost function $C'(q)$, calculate $C'(10)$, and interpret its units and meaning.

c) Find the equation of the tangent line to the graph of C at $q = 10$.

d) Use the tangent line to estimate the cost of producing 11 units. Compare it with the exact value.

Problem 3 — (Product rule · chain rule · exponential functions)

The price, in euros per unit, at which a firm can sell q units is $p(q) = 80e^{-0.02q}$. Revenue is $R(q) = q \cdot p(q)$.

a) Calculate $p'(q)$ and explain the meaning of its sign.

b) Calculate $R'(q)$ and simplify the result.

c) Calculate and interpret $R'(25)$.

d) Find the tangent-line approximation to R at $q = 25$ and use it to estimate $R(27)$. Compare with the exact value.

e) Calculate $R'(50)$. What is special about $q = 50$?

Problem 4 — (Differentiability of a piecewise function)

Let

$$f(x) = \begin{cases} x^2 & \text{if } x < 2 \\ 4x - 4 & \text{if } 2 \leq x \leq 4 \\ 12 + \ln(x - 3) & \text{if } x > 4 \end{cases}$$

a) Show that f is continuous at $x = 2$ and at $x = 4$.

b) Find, if they exist, $f'(0)$, $f'(2)$, $f'(4)$ and $f'(5)$. Justify with one-sided derivatives.

Problem 5 — (Composition · chain rule interpretation)

Let $P(w)$ be the number of units a factory produces per day with w workers, and let $R(u)$ be the daily revenue, in euros, from selling u units.

a) Interpret the function $b(w) = (R \circ P)(w)$ and its derivative $b'(w)$, including units.

b) You know that $P(10) = 500$, $P'(10) = 40$ and $R'(500) = 12$. Calculate $b'(10)$ and interpret it.

Problem 6 — (Rules of differentiation)

Calculate and simplify:

a) $f(x) = x^2 e^{1/x}$

b) $f(x) = \ln\bigl(x^2 + 1\bigr)$

c) $f(x) = \dfrac{x^2 + 1}{x - 1}$

d) $f(x) = e^{x^2} \ln x$

e) $f(x) = \sqrt{3x^2 + 1}$

f) $f(x) = \dfrac{e^{2x}}{x}$

Then find the tangent line to $g(x) = \ln(x^2 + 1)$ at $x = 1$.

Problem 7 — (Marginal cost vs average cost)

A company has total cost $C(x) = 2000 + 15x + 0.2x^2$ euros, where $x > 0$ is the number of units. Its average cost is $A(x) = \dfrac{C(x)}{x}$.

a) Find $C'(x)$ and $A'(x)$ and simplify.

b) Calculate $A'(50)$ and $A'(150)$ and interpret their signs.

c) Calculate $C'(100)$ and $A(100)$. What do you notice?

Problem 8 — (Second derivatives · interpretation)

Demand for a concert ticket is $q(p) = 5000e^{-0.04p}$, where p is the price in euros. Revenue is $I(p) = p\,q(p)$.

a) Find $q'(p)$ and $q''(p)$. Calculate $q'(25)$ and $q''(25)$ and explain what they say about demand around €25.

b) Find $I'(p)$ and simplify. Calculate $I'(25)$ and interpret it.

Problems · Topic 2.3

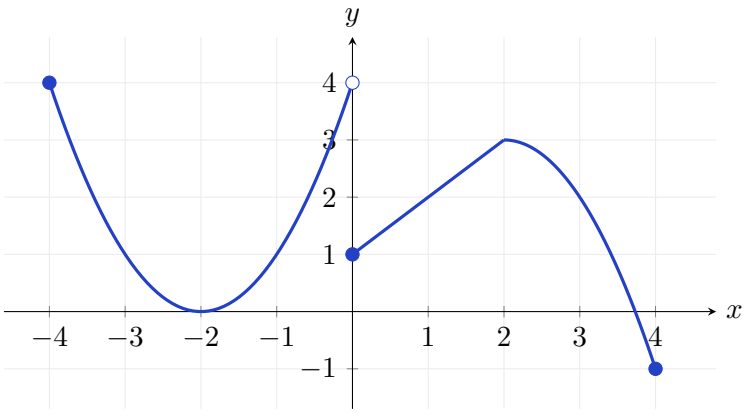
UNIT 2. ONE-VARIABLE CALCULUS

TOPIC 2.3. FUNCTION ANALYSIS AND OPTIMIZATION

Problems · Answers at the end of the book

Problem 1 — (Increasing and decreasing · extreme points from a graph)

The figure shows the graph of $g(x)$ on its domain $[-4, 4]$. You know that $g(-4) = 4$ and $g(0) = 1$.



Determine the intervals of continuity and differentiability, the increasing and decreasing intervals, and all local and global maxima and minima.

Problem 2 — (Increasing and decreasing · extreme points · concavity)

Let $f(x) = 2x^3 - 3x^2 - 12x + 5$.

a) Find the intervals where f is increasing and decreasing, and its local extrema.

b) Find the intervals of concavity and the inflection point.

c) Find the global maximum and minimum of f on $[-2, 3]$.

Problem 3 — (Economic examples · optimization with price as the variable)

The number of units sold is $x = 100 - p$, where p is the selling price in euros. The total cost of producing and selling x units is $C(x) = 0.5x^2 + 10x + 200$.

a) Determine the profit $\pi(p)$ as a function of the price. Define its mathematical and contextual domain.

b) Find the increasing and decreasing intervals of π, the optimal price, the number of units sold, and the maximum profit.

c) Show that this is the global maximum on the contextual domain.

Problem 4 — (Piecewise model · continuity · differentiability · optimization)

Weekly orders $f(x)$ of a protein bar, as a function of the grams x of nuts in each bar, are modelled by

$$f(x) = \begin{cases} 0 & \text{if } 0 \le x < 10 \\ 8x & \text{if } 10 \le x < 40 \\ x(16 - 0.2x) & \text{if } 40 \le x \le 80 \end{cases}$$

Justify your answers using the equation of the function.

a) Calculate, if possible, $\lim_{x\to 10} f(x)$. If not, explain why.

b) Is f continuous at $x = 10$?

c) Is f differentiable at $x = 40$?

d) Calculate $f'(60)$ and write one sentence interpreting it in context.

e) What is the optimal amount of nuts to maximize orders? How many orders is that?

Problem 5 — (Exponential model · constrained optimization)

Monthly profit, in thousands of euros, is $P(x) = 80 - 100e^{-0.05x} - 2x$, where x is the monthly advertising spend in thousands of euros and $0 \le x \le 50$.

a) Find $P'(x)$ and all stationary points in the contextual domain.

b) Classify them with $P''(x)$.

c) Find the global maximum and minimum and interpret them in context.

Problem 6 — (Minimizing average cost)

For the cost $C(x) = 2000 + 15x + 0.2x^2$ of Topic 2.2, Problem 7, the average cost is $A(x) = \dfrac{2000}{x} + 15 + 0.2x$, $x > 0$.

a) Find the production level that minimizes average cost and the minimum average cost.

b) Justify with $A''(x)$ that it is a minimum.

c) Show that at this production level marginal cost equals average cost.

Problem 7 — (Asymptotes · extrema · inflection)

Let $f(x) = (x - 1)e^x$.

a) Find the local extrema of f.

b) Find the intervals of concavity and the inflection point.

c) Calculate $\lim_{x\to -\infty} f(x)$. Does f have a horizontal asymptote?

Problem 8 — (Piecewise function · kinks)

Let $f(x) = (x - 1)^2$ if $x \le 3$ and $f(x) = 7 - x$ if $x > 3$.

a) Study the continuity and differentiability of f.

b) Find the increasing and decreasing intervals.

c) Find all local and global extrema, if any. Explain why one of the local maxima is not a stationary point.

Problems · Topic 2.4

UNIT 2. ONE-VARIABLE CALCULUS

TOPIC 2.4. TAYLOR APPROXIMATIONS

Problems · Answers at the end of the book

Problem 1 — (Taylor polynomial · concavity)
Given $f(x) = x\,e^{-x}$:

a) Find the third-degree Taylor polynomial of f at $x = 1$.

b) Use it to decide whether f is concave upwards or downwards at $x = 1$, and sketch f near $x = 1$ together with its tangent line. What kind of point is $x = 1$?

Problem 2 — (Reading derivatives from a Taylor polynomial)
The sixth-degree Taylor polynomial of g at $x = 2$ is $P(x) = 1 - 2(x-2) + 3(x-2)^6$.

a) Find the values of the derivatives of g (from the first to the sixth) at $x = 2$.

b) Analyze the monotonicity and the concavity of g near $x = 2$, and sketch g with its tangent line.

Problem 3 — (Taylor polynomial around a point $a \neq 0$ · estimation)
Let $f(x) = \ln x$.

a) Find $P_3(x)$ around $x = e$.

b) Use it to estimate $\ln(2.8)$ and compare with the calculator value.

Problem 4 — (Piecewise function · Taylor approximation on one piece)
Let

$$f(x) = \begin{cases} x^2 - 3 & \text{if } x \leq 1 \\ 2\sqrt{x} & \text{if } x > 1 \end{cases}$$

a) Calculate the third-order Taylor approximation of f around $x = 4$.

b) Use it to estimate $f(4.2)$, and sketch f and its tangent line near $x = 4$.

Problem 5 — (Business application · stationary point)
Revenue as a function of price is $R(p) = 100\,p\,e^{-0.1p}$ euros.

a) Find the third-order Taylor polynomial of R around $p = 10$.

b) What does it tell you about revenue around $p = 10$? Sketch R near $p = 10$.

c) Use the polynomial to estimate $R(11)$.

Problem 6 — (Approximating a root)
Let $f(x) = \sqrt[3]{x}$. Find $P_2(x)$ around $x = 8$ and use it to estimate $\sqrt[3]{8.3}$.

Problem 7 — (Building Taylor polynomials from known ones)

a) Using $e^u \approx 1 + u + \dfrac{u^2}{2}$, write a fourth-order polynomial approximation of $f(x) = e^{-x^2}$ around $x = 0$.

b) Check that it gives the correct value of $f''(0)$ by differentiating f.

c) A firm's third-order Taylor approximation of sales around the current price has a positive third-degree term. Is it correct to conclude that the firm should raise prices? Explain.

Problems · Topic 2.5

UNIT 2. ONE-VARIABLE CALCULUS

TOPIC 2.5. INTEGRATION

Problems · Answers at the end of the book

Problem 1 — (Indefinite integrals · initial condition)

The marginal cost of a product is $\dfrac{dC}{dq} = \dfrac{6}{\sqrt{q}}$ euros per unit. Producing 100 units costs €500. Calculate the total cost of producing 400 units.

Problem 2 — (Indefinite and definite integrals · economic application)

The marginal cost, in euros per unit, when x units are produced is $m(x) = 20e^{0.02x}$, $x \geq 0$.

a) How much does total cost increase when production rises from 0 to 50 units?

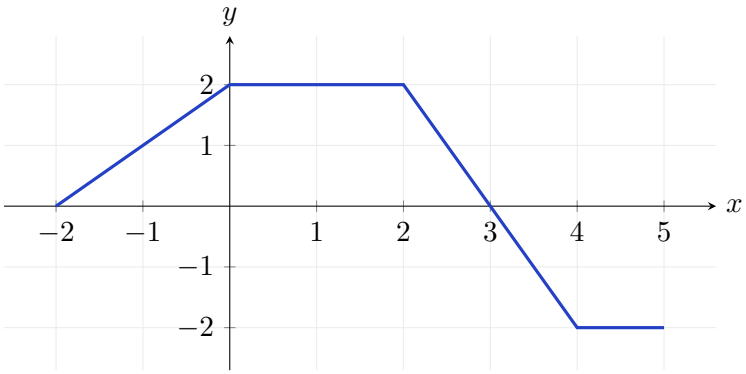
b) Find the total cost function $C(x)$ knowing that the fixed cost is €2 000.

Problem 3 — (Definite integral with a parameter)

The definite integral of $f(x) = x^2 + ax + 2$ on $[0, 3]$ is 18. Calculate a.

Problem 4 — (Area and definite integrals from a graph)

The graph of $f(x)$ is made of straight segments:



a) Calculate $\displaystyle\int_{-2}^{5} f(x)\,dx$ and $\displaystyle\int_{0}^{4} f(x)\,dx$.

b) Calculate the total area between the graph of f and the x-axis on $[-2, 5]$. Why is it different from the first integral in (a)?

Problem 5 — (Area of a region)

a) Calculate the area of the region bounded by $y = e^x$, $y = e$ and $x = 0$.

b) Calculate the area of the region bounded by $y = x^2$ and $y = 4$.

Problem 6 — (Properties of definite integrals)

Calculate, splitting the interval where necessary:

a) $\displaystyle\int_{-1}^{3} |x - 1|\,dx$

b) $\displaystyle\int_0^4 \left\lfloor \frac{x}{2} \right\rfloor dx$

Problem 7 — (Antiderivatives · algebra with F')
If $f(x) = F'(x)$ and $g(x) = 2F'(x) + 3x^2$, determine all functions $H(x)$ such that

$$H(x) = \int \big(4f(x) - 2g(x)\big)\, dx.$$

Problem 8 — (Density functions)
Let $f(x) = kx^2$ for $0 \le x \le 3$ and $f(x) = 0$ otherwise.

a) Determine k so that f is a density function.

b) Calculate $\displaystyle\int_0^1 f(x)\, dx$, the probability that the variable is at most 1.

c) Compute the average value of f on $[0, 3]$.

Problem 9 — (Marginal revenue and marginal cost)
A firm has marginal revenue $R'(x) = 100 - 0.4x$ and marginal cost $C'(x) = 20 + 0.2x$, in euros per unit. $R(0) = 0$ and the fixed cost is €1 000.

a) Find $R(x)$ and the quantity that maximizes revenue.

b) Find $C(x)$ and the profit function $\pi(x)$.

c) Find the quantity that maximizes profit and the maximum profit.

Problem 10 — (Average value)
Daily sales, in units, t days after a promotion starts are $S(t) = 50 + 20e^{-0.1t}$. Find the average daily sales over the first 10 days. Give the exact value and a decimal.

Past-paper style exam C

UNIT 2. ONE-VARIABLE CALCULUS

PAST-PAPER STYLE EXAM C · TOPICS 2.1 – 2.5

90 minutes · Formula sheet allowed

Problem 1 — (Continuity and differentiability · 2.1–2.2 · 25 points)

Let

$$f(x) = \begin{cases} e^x + a & \text{if } x < 0 \\ x^2 + bx + 2 & \text{if } 0 \le x \le 2 \\ \ln(x-1) + c & \text{if } x > 2 \end{cases}$$

a) Find a, b and c so that f is continuous on $\mathbb{R}$ and differentiable at $x = 0$.

b) With those values, is f differentiable at $x = 2$? Justify.

c) Calculate $\lim_{x \to -\infty} f(x)$ and $\lim_{x \to +\infty} f(x)$, and give the asymptotes of f.

Problem 2 — (Optimization · 2.3 · 25 points)

A firm sells its product at a fixed price of €100 per unit. Its total cost of producing q units is

$$C(q) = \frac{q^3}{3} - 12q^2 + 180q + 300, \qquad 0 \le q \le 30.$$

a) Write the profit function $\pi(q)$.

b) Find the stationary points of π and classify them with π''.

c) Find the global maximum and minimum of profit on $[0, 30]$ and interpret.

d) Find the inflection point of π and interpret it as a change in marginal profit.

Problem 3 — (Taylor approximation · 2.4 · 25 points)

Let $f(x) = \ln(1 + 2x)$.

a) Find its domain and the third-order Taylor polynomial around $x = 0$.

b) Use it to estimate $\ln(1.2)$ and compare with the calculator value.

c) Sketch f and its tangent line near $x = 0$, justifying the position of the curve.

Problem 4 — (Integration · 2.5 · 25 points)

A firm has marginal revenue $R'(q) = 200 - 0.8q$ and marginal cost $C'(q) = 40 + 0.2q$, in euros per unit. The fixed cost is €3 000 and $R(0) = 0$.

a) Find $R(q)$ and $C(q)$.

b) Find the output that maximizes profit.

c) Calculate the maximum profit as an integral of marginal profit.

d) By how much does revenue increase when output rises from 100 to 160 units?

Past-paper style exam D

UNIT 2. ONE-VARIABLE CALCULUS

PAST-PAPER STYLE EXAM D · TOPICS 2.1 – 2.5

90 minutes · Formula sheet allowed

Problem 1 — (Limits · derivatives · tangent line · 2.1–2.2 · 25 points)

The cost, in euros, of removing $x\%$ of a pollutant from a lake is $C(x) = \dfrac{8000\,x}{120 - x}$.

a) Find the mathematical and contextual domain of C. Can the pollutant be removed completely? At what cost?

b) Prove that C is increasing on its contextual domain.

c) Calculate $C'(60)$ and interpret it.

d) Use the tangent line at $x = 60$ to estimate $C(61)$, and compare with the exact value.

Problem 2 — (Function analysis · 2.3 · 25 points)

Let $f(x) = x^3 - 3x$ on $[-2, 3]$.

a) Find the increasing and decreasing intervals and the local extrema.

b) Find the inflection point.

c) Find the global maximum and minimum on $[-2, 3]$.

Problem 3 — (Taylor polynomials · 2.4 · 25 points)

a) The fourth-degree Taylor polynomial of f at $x = -1$ is $P_4(x) = 5 - 3(x+1)^2 + 2(x+1)^4$. Find $f(-1), \ldots, f^{(4)}(-1)$, the tangent line, and the type of point $x = -1$ is. Sketch f nearby.

b) The third-degree Taylor polynomial of h at $x = 2$ is $P_3(x) = -1 + 2(x-2) - \dfrac{1}{2}(x-2)^3$. Find $h'''(2)$ and describe h near $x = 2$ relative to its tangent line.

Problem 4 — (Density function · area · 2.5 · 25 points)

a) Let $f(x) = k(4 - x^2)$ for $-2 \le x \le 2$ and 0 otherwise. Find k so that f is a density function, and calculate $\displaystyle\int_{-1}^{1} f(x)\,dx$.

b) Calculate the area of the region bounded by $y = x^2$ and $y = x + 2$.

Problem answers · Topic 2.1

1. Equilibrium means $q_{sup} = q_{dem}$:

$$3p^2 = \frac{375}{p} \iff 3p^3 = 375 \iff p^3 = 125 \iff p = 5.$$

At $p = 5$: $q_{sup} = 3 \cdot 25 = 75$ and $q_{dem} = \dfrac{375}{5} = 75$. Equilibrium price 5 monetary units, quantity 75 units.

2. (i) $+\infty$: the dashed vertical line $x = -1$ is a vertical asymptote and the curve rises on its left.
(ii) -1: from the right the middle piece approaches the empty dot $(-1, -1)$.
(iii) $\lim\limits_{x\to 2^-} f(x) = 2$, so the limit is $2 + 1 = 3$.
(iv) Does not exist: $\lim\limits_{x\to 2^-} f(x) = 2 \neq 1 = \lim\limits_{x\to 2^+} f(x)$.
(v) 1: horizontal asymptote $y = 1$ on the left.
(vi) $\dfrac{1}{\lim\limits_{x\to+\infty} f(x)} = \dfrac{1}{3}$: horizontal asymptote $y = 3$ on the right.
Discontinuities: $x = -1$ is an **infinite (asymptotic)** discontinuity; $x = 2$ is a **jump** discontinuity.

3. a) Mathematical domain $\mathbb{R} \setminus \{100\}$. Since x is a percentage that can be removed, the contextual domain is $[0, 100)$.
b) $C(50) = \dfrac{20\,000 \cdot 50}{50} = 20\,000$.
c) $\dfrac{20\,000x}{100 - x} = 80\,000 \iff 20\,000x = 8\,000\,000 - 80\,000x \iff 100\,000x = 8\,000\,000 \iff x = 80\%$.
With €180 000: $200\,000x = 18\,000\,000 \iff x = 90\%$. Note that the extra €100 000 only removes 10 more percentage points.
d) $\lim\limits_{x\to 100^-} \dfrac{20\,000x}{100 - x} = \dfrac{2\,000\,000}{0^+} = +\infty$: removing all the plastic would have an unlimited cost, so it is impossible in practice.

4. a) At $x = 1$: $\lim\limits_{x\to 1^-} (x^2 + 1) = 2$ and $\lim\limits_{x\to 1^+} (ax + 3) = f(1) = a + 3$. Continuity: $2 = a + 3 \Rightarrow a = -1$.
At $x = 4$ (with $a = -1$): $\lim\limits_{x\to 4^-} (-x + 3) = f(4) = -1$ and $\lim\limits_{x\to 4^+} \big(\ln(x - 3) + b\big) = \ln 1 + b = b$.
Continuity: $b = -1$.
So f is continuous at $x = 1$ and $x = 4$ for $a = -1$, $b = -1$.
b) With $a = 1$, $b = 0$: at $x = 1$ the left limit is 2 and the right limit is $f(1) = 4$, so there is a **jump**. At $x = 4$ the left limit is $f(4) = 7$ and the right limit is 0, another **jump**. Each piece is continuous on its own interval ($\ln(x - 3)$ is defined for $x > 4$), so f is continuous on $\mathbb{R} \setminus \{1, 4\}$.

5. a) $h(x) = f\big(g(x)\big) = \sqrt{\dfrac{1}{x - 5} - 2}$ $\qquad$ $m(x) = g\big(f(x)\big) = \dfrac{1}{\sqrt{x - 2} - 5}$.
b) h: we need $x \neq 5$ and $\dfrac{1}{x - 5} \geq 2$. If $x < 5$ the left side is negative, impossible. If $x > 5$: $1 \geq 2(x - 5) \iff x \leq 5.5$. So $D(h) = (5,\ 5.5]$.
m: we need $x \geq 2$ and $\sqrt{x - 2} \neq 5 \iff x \neq 27$. So $D(m) = [2, 27) \cup (27, +\infty)$.

6. a) $p(q) > 0 \iff 90 - 0.5q > 0 \iff q < 180$, and $q \geq 0$: $D_c = [0, 180)$.
b) $I(q) = q(90 - 0.5q) = 90q - 0.5q^2$. $\pi(q) = 90q - 0.5q^2 - 600 - 20q - 0.5q^2 = -q^2 + 70q - 600$.
c) $-q^2 + 70q - 600 = 0 \iff q^2 - 70q + 600 = 0 \iff q = \dfrac{70 \pm \sqrt{4900 - 2400}}{2} = \dfrac{70 \pm 50}{2}$, so $q = 10$ or $q = 60$. The parabola opens downwards, so the firm makes a profit for $10 < q < 60$ units (maximum at $q = 35$, $\pi = 625$).
d) $A(q) = \dfrac{600}{q} + 20 + 0.5q$, so $\lim\limits_{q\to 0^+} A(q) = \dfrac{600}{0^+} + 20 = +\infty$. With very few units, the fixed cost of €600 is spread over almost nothing, so the cost per unit explodes.

7. a) $N(0) = \dfrac{2000}{1+9} = 200$ users.

b) $e^{-0.5t} \to e^{-\infty} = 0$, so $\lim\limits_{t\to+\infty} N(t) = \dfrac{2000}{1+0} = 2000$. In the long run the app approaches 2 000 users (market saturation). $y = 2000$ is a horizontal asymptote.

c) $\lim\limits_{t\to+\infty} M(t) = 500(1-0) = 500$, so $\lim\limits_{t\to+\infty} \big(N(t) - M(t)\big) = 2000 - 500 = 1500$.

8. a) $P(0.5) = 3 + 2\cdot 0 = 3$, $P(2.5) = 3 + 2\cdot 2 = 7$, $P(3) = 3 + 2\cdot 3 = 9$.

b) $\lfloor t \rfloor \in \{0,1,2,3,4,5\}$ for $0 < t \le 5$, so $R(P) = \{3,5,7,9,11,13\}$.

c) At $t = 1,2,3,4$: e.g. at $t = 2$, $\lim\limits_{t\to 2^-} P(t) = 3+2\cdot 1 = 5$ and $\lim\limits_{t\to 2^+} P(t) = 7$. Both finite and different: **jump** discontinuities at $t = 1,2,3,4$.

d) Examples: quantity discounts (a lower unit price from 100 units on), tax brackets, a fixed cost when a second shift or machine is needed, shipping fees by weight band.

Problem answers · Topic 2.2

1. a) Quotient rule:

$$C'(x) = \frac{20\,000(100-x) - 20\,000x\cdot(-1)}{(100-x)^2} = \frac{2\,000\,000}{(100-x)^2} > 0$$

for every x in $[0, 100)$, so C is increasing on its whole contextual domain.

b) $C'(50) = \dfrac{2\,000\,000}{2500} = 800$ €/percentage point: at 50% cleaned, one more percentage point costs about €800.

$C'(90) = \dfrac{2\,000\,000}{100} = 20\,000$ €/percentage point: at 90%, one more point costs about €20 000. The last percentages are by far the most expensive.

2. a) $C(10) = 10 - 60 + 300 + 400 = 650$ and $C(20) = 80 - 240 + 600 + 400 = 840$:

$$\frac{C(20) - C(10)}{20 - 10} = \frac{190}{10} = 19 \text{ per unit.}$$

b) $C'(q) = 0.03q^2 - 1.2q + 30$, $C'(10) = 3 - 12 + 30 = 21$ euros per unit: at 10 units, total cost increases at about €21 per extra unit.

c) $L(q) = 650 + 21(q - 10) = 21q + 440$.

d) $L(11) = 671$. Exact: $C(11) = 13.31 - 72.6 + 330 + 400 = 670.71$. The tangent line overestimates by €0.29.

3. a) $p'(q) = 80e^{-0.02q}\cdot(-0.02) = -1.6e^{-0.02q} < 0$: the more units the firm wants to sell, the lower the price must be.

b) Product rule: $R'(q) = 1\cdot 80e^{-0.02q} + q\cdot(-1.6e^{-0.02q}) = 80e^{-0.02q}(1 - 0.02q)$.

c) $R'(25) = 80e^{-0.5}\cdot 0.5 = 40e^{-0.5} \approx 24.26$ per unit: at 25 units, selling one more unit adds about €24.26 of revenue.

d) $R(25) = 2000e^{-0.5} \approx 1213.06$, so $L(q) = 1213.06 + 24.26(q - 25)$ and $R(27) \approx L(27) = 1213.06 + 2\cdot 24.26 = 1261.58$. Exact: $R(27) = 2160e^{-0.54} \approx 1258.74$. The estimate is about €2.85 too high.

e) $R'(50) = 80e^{-1}(1-1) = 0$: $q = 50$ is a stationary point of revenue (it is where revenue is highest, since $R' > 0$ before and $R' < 0$ after).

4. a) At $x = 2$: left $2^2 = 4$, right and value $4\cdot 2 - 4 = 4$. At $x = 4$: left and value $4\cdot 4 - 4 = 12$, right $12 + \ln 1 = 12$. Continuous at both.

b) $f'(0) = 2\cdot 0 = 0$ (first piece).

At $x = 2$: left derivative $2x \to 4$, right derivative 4. Equal, so $f'(2) = 4$: f **is** differentiable at 2 (the pieces join smoothly).

At $x = 4$: left derivative 4, right derivative $\dfrac{1}{x-3} \to 1$. Different, so $f'(4)$ does not exist (a kink).

$f'(5) = \dfrac{1}{5-3} = \dfrac{1}{2}$.

5. a) $b(w) = R\big(P(w)\big)$ is the daily revenue, in euros, when the factory has w workers. $b'(w)$ is the rate of change of daily revenue per additional worker, in euros per worker.
b) Chain rule: $b'(10) = R'\big(P(10)\big) \cdot P'(10) = R'(500) \cdot 40 = 12 \cdot 40 = 480$. With 10 workers, one more worker adds about €480 of daily revenue.

6. a) $2x\,e^{1/x} + x^2 e^{1/x} \cdot \left(-\dfrac{1}{x^2}\right) = e^{1/x}(2x-1)$

b) $\dfrac{2x}{x^2+1}$

c) $\dfrac{2x(x-1)-(x^2+1)}{(x-1)^2} = \dfrac{x^2-2x-1}{(x-1)^2}$

d) $2x\,e^{x^2}\ln x + e^{x^2}\cdot\dfrac{1}{x} = e^{x^2}\left(2x\ln x + \dfrac{1}{x}\right)$

e) $\dfrac{1}{2\sqrt{3x^2+1}}\cdot 6x = \dfrac{3x}{\sqrt{3x^2+1}}$

f) $\dfrac{2e^{2x}\cdot x - e^{2x}}{x^2} = \dfrac{e^{2x}(2x-1)}{x^2}$

Tangent: $g(1) = \ln 2$, $g'(1) = \dfrac{2}{2} = 1$, so $L(x) = \ln 2 + (x-1)$.

7. a) $C'(x) = 15 + 0.4x$. $A(x) = \dfrac{2000}{x} + 15 + 0.2x$, so $A'(x) = -\dfrac{2000}{x^2} + 0.2$.
b) $A'(50) = -0.8 + 0.2 = -0.6$: around 50 units, average cost falls by about €0.60 per extra unit.
$A'(150) = -0.089 + 0.2 \approx 0.11$: around 150 units, average cost rises by about €0.11 per extra unit.
c) $C'(100) = 15 + 40 = 55$ and $A(100) = 20 + 15 + 20 = 55$. They are equal. (Topic 2.3, Problem 6 shows this is exactly where average cost is lowest.)

8. a) $q'(p) = -200e^{-0.04p}$ and $q''(p) = 8e^{-0.04p}$.
$q'(25) = -200e^{-1} \approx -73.6$: at €25, a €1 increase loses about 74 tickets.
$q''(25) = 8e^{-1} \approx 2.94 > 0$: q' is increasing (becoming less negative), so each further euro loses fewer tickets than the one before.
b) $I'(p) = 5000e^{-0.04p} + p(-200e^{-0.04p}) = 5000e^{-0.04p}(1 - 0.04p)$.
$I'(25) = 5000e^{-1}(1-1) = 0$: €25 is a stationary point of revenue. Revenue grows for $p < 25$ and falls for $p > 25$, so €25 maximizes revenue.

Problem answers · Topic 2.3

1. Continuity: g is continuous on $[-4,4] \setminus \{0\}$; at $x = 0$ there is a jump ($\lim\limits_{x\to 0^-} g = 4$, $\lim\limits_{x\to 0^+} g = 1 = g(0)$).
Differentiability: not at $x = 0$ (not even continuous) and not at $x = 2$ (corner: slope 1 on the left, slope 0 on the right). Differentiable on $(-4,0) \cup (0,2) \cup (2,4)$.
Decreasing on $(-4,-2)$ and $(2,4)$; increasing on $(-2,0)$ and $(0,2)$.
Local maxima: $x = -4$ (endpoint, $g = 4$) and $x = 2$ ($g = 3$).
Local minima: $x = -2$ ($g = 0$), $x = 0$ ($g(0) = 1$: values just left are close to 4 and values just right are above 1), and $x = 4$ (endpoint, $g = -1$).
Global maximum 4 at $x = -4$ (the value 4 approached at 0^- is never reached). Global minimum -1 at $x = 4$.

2. a) $f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$. $f' > 0$ on $(-\infty,-1)$ and $(2,+\infty)$: increasing. $f' < 0$ on $(-1,2)$: decreasing.
Local max at $x = -1$: $f(-1) = 12$. Local min at $x = 2$: $f(2) = -15$.
b) $f''(x) = 12x - 6 = 0 \iff x = \dfrac{1}{2}$, and f'' changes sign. Concave down on $\left(-\infty, \dfrac{1}{2}\right)$, up on

$\left(\frac{1}{2}, +\infty\right)$. Inflection point $\left(\frac{1}{2}, -\frac{3}{2}\right)$.
c) Candidates: $f(-2) = 1$, $f(-1) = 12$, $f(2) = -15$, $f(3) = -4$. Global max 12 at $x = -1$, global min -15 at $x = 2$.

3. a) Income: $I(p) = p(100 - p) = 100p - p^2$. Cost as a function of price (composition):

$$C(p) = 0.5(100 - p)^2 + 10(100 - p) + 200.$$

$$\pi(p) = 100p - p^2 - \left[0.5(100 - p)^2 + 10(100 - p) + 200\right] = -1.5p^2 + 210p - 6200.$$

Mathematical domain $\mathbb{R}$. Units and price cannot be negative, so $0 \le p \le 100$: $D_c = [0, 100]$.
b) $\pi'(p) = -3p + 210$. Positive for $p < 70$ (increasing), negative for $p > 70$ (decreasing). Optimal price $p = 70$, units sold $x = 30$, maximum profit $\pi(70) = -7350 + 14\,700 - 6200 = 1150$.
c) $\pi'' = -3 < 0$, and the endpoints give $\pi(0) = -6200$ and $\pi(100) = -200$, both lower. So $p = 70$ is the global maximum on $[0, 100]$.

4. a) $\lim_{x \to 10^-} f(x) = 0$ and $\lim_{x \to 10^+} f(x) = 8 \cdot 10 = 80$. The one-sided limits differ, so the limit does not exist.
b) No: the limit at $x = 10$ does not exist, so f is not continuous there (jump).
c) $f'(x) = 8$ on $(10, 40)$ and $f'(x) = 16 - 0.4x$ on $(40, 80)$. $\lim_{x \to 40^-} f'(x) = 8 \neq 0 = \lim_{x \to 40^+} f'(x)$. Not differentiable at $x = 40$ (it is continuous there: $8 \cdot 40 = 320 = 40 \cdot 8$).
d) $f'(60) = 16 - 24 = -8$: with 60 g of nuts, adding one more gram reduces weekly orders by about 8.
e) f increases on $(10, 40)$ and decreases on $(40, 80)$, so the maximum is at the kink $x = 40$: 40 g of nuts, $f(40) = 320$ orders per week.

5. a) $P'(x) = -100e^{-0.05x}(-0.05) - 2 = 5e^{-0.05x} - 2$. $P'(x) = 0 \iff e^{-0.05x} = 0.4 \iff x = 20 \ln 2.5 \approx 18.33$, inside $[0, 50]$.
b) $P''(x) = -0.25e^{-0.05x} < 0$: local maximum.
c) $P(0) = -20$, $P(18.33) = 80 - 40 - 36.65 \approx 3.35$, $P(50) = -20 - 100e^{-2.5} \approx -28.21$. Spend about €18 330 a month on advertising for a maximum monthly profit of about €3 350. The global minimum is at $x = 50$ (overspending).

6. a) $A'(x) = -\frac{2000}{x^2} + 0.2 = 0 \iff x^2 = 10\,000 \iff x = 100$. $A(100) = 20 + 15 + 20 = 55$ per unit.
b) $A''(x) = \frac{4000}{x^3} > 0$ for $x > 0$: minimum.
c) $C'(100) = 15 + 0.4 \cdot 100 = 55 = A(100)$.

7. a) $f'(x) = e^x + (x - 1)e^x = x\,e^x = 0 \iff x = 0$. $f' < 0$ for $x < 0$, $f' > 0$ for $x > 0$: local (and global) minimum at $(0, -1)$.
b) $f''(x) = e^x + xe^x = (x + 1)e^x$. Concave down on $(-\infty, -1)$, up on $(-1, +\infty)$. Inflection point $\left(-1, -\frac{2}{e}\right)$.
c) As $x \to -\infty$, $e^x \to 0$ much faster than $x - 1 \to -\infty$ grows, so $f(x) \to 0$: horizontal asymptote $y = 0$ on the left.

8. a) At $x = 3$: left $(3 - 1)^2 = 4$, right $7 - 3 = 4$, $f(3) = 4$: continuous on $\mathbb{R}$. Slopes at 3: left $2(x - 1) \to 4$, right -1. Not differentiable at $x = 3$.
b) $f'(x) = 2(x - 1)$ for $x < 3$ and -1 for $x > 3$. Decreasing on $(-\infty, 1)$ and $(3, +\infty)$; increasing on $(1, 3)$.
c) Global (and local) minimum $f(1) = 0$. Local maximum at $x = 3$, $f(3) = 4$; not global, since e.g. $f(-2) = 9 > 4$; no global maximum ($f \to +\infty$ as $x \to -\infty$). The maximum at $x = 3$ is at a kink, where f' does not exist, so it is not a stationary point.

Problem answers · Topic 2.4

1. a) $f(x) = xe^{-x} \Rightarrow f(1) = \dfrac{1}{e}$
$f'(x) = (1-x)e^{-x} \Rightarrow f'(1) = 0$
$f''(x) = (x-2)e^{-x} \Rightarrow f''(1) = -\dfrac{1}{e}$
$f'''(x) = (3-x)e^{-x} \Rightarrow f'''(1) = \dfrac{2}{e}$

$$P_3(x) = \frac{1}{e} - \frac{(x-1)^2}{2e} + \frac{(x-1)^3}{3e}.$$

b) Tangent line: $y = \dfrac{1}{e}$ (horizontal). The first term after it is $-\dfrac{(x-1)^2}{2e} \leq 0$: an even power with a negative coefficient, so f is **below** its tangent on both sides near $x = 1$: concave downwards, and $x = 1$ is a local maximum.

2. a) Compare with $P(x) = g(2) + g'(2)(x-2) + \cdots + \dfrac{g^{(6)}(2)}{720}(x-2)^6$:

$g(2) = 1$, $g'(2) = -2$, $g''(2) = g'''(2) = g^{(4)}(2) = g^{(5)}(2) = 0$, and $\dfrac{g^{(6)}(2)}{720} = 3 \Rightarrow g^{(6)}(2) = 2160$.
b) $g'(2) = -2 < 0$, so g is decreasing near $x = 2$ (also $P'(x) = -2 + 18(x-2)^5 \approx -2$ there). Tangent: $y = 1 - 2(x-2)$. $P - P_1 = 3(x-2)^6 \geq 0$, an even power with a positive coefficient: g lies **above** its tangent on both sides, so it is concave upwards near $x = 2$.

3. a) $f(e) = 1$, $f'(e) = \dfrac{1}{e}$, $f''(x) = -\dfrac{1}{x^2} \Rightarrow f''(e) = -\dfrac{1}{e^2}$, $f'''(x) = \dfrac{2}{x^3} \Rightarrow f'''(e) = \dfrac{2}{e^3}$:

$$P_3(x) = 1 + \frac{x-e}{e} - \frac{(x-e)^2}{2e^2} + \frac{(x-e)^3}{3e^3}.$$

b) With $x = 2.8$: $P_3(2.8) \approx 1.029620$, and $\ln(2.8) \approx 1.029619$. Correct to 5 decimals.

4. a) Near $x = 4$ we use $f(x) = 2\sqrt{x} = 2x^{1/2}$ (since $4 > 1$):
$f(4) = 4$, $f'(x) = x^{-1/2} \Rightarrow f'(4) = \dfrac{1}{2}$, $f''(x) = -\dfrac{1}{2}x^{-3/2} \Rightarrow f''(4) = -\dfrac{1}{16}$, $f'''(x) = \dfrac{3}{4}x^{-5/2} \Rightarrow f'''(4) = \dfrac{3}{128}$.

$$P_3(x) = 4 + \frac{1}{2}(x-4) - \frac{1}{32}(x-4)^2 + \frac{1}{256}(x-4)^3.$$

b) $f(4.2) \approx 4 + 0.1 - 0.00125 + 0.00003 = 4.09878$ (exact $2\sqrt{4.2} \approx 4.09878$). Tangent: $y = 4 + \dfrac{1}{2}(x-4)$. The first term after it, $-\dfrac{1}{32}(x-4)^2$, is negative: f lies below its tangent, concave downwards.

5. a) $R(p) = 100pe^{-0.1p}$: $R(10) = \dfrac{1000}{e}$
$R'(p) = 100e^{-0.1p}(1 - 0.1p) \Rightarrow R'(10) = 0$
$R''(p) = 100e^{-0.1p}(0.01p - 0.2) \Rightarrow R''(10) = -\dfrac{10}{e}$
$R'''(p) = 100e^{-0.1p}(0.03 - 0.001p) \Rightarrow R'''(10) = \dfrac{2}{e}$

$$P_3(p) = \frac{1000}{e} - \frac{5}{e}(p-10)^2 + \frac{1}{3e}(p-10)^3.$$

b) The tangent is horizontal ($R'(10) = 0$) and the first term after it, $-\dfrac{5}{e}(p-10)^2$, is negative: revenue lies below its tangent on both sides, so $p = 10$ is a local maximum of revenue (about €367.88).
c) $R(11) \approx \dfrac{1000 - 5 + \frac{1}{3}}{e} \approx 366.16$ (exact $1100e^{-1.1} \approx 366.16$).

6. $f(8)=2$, $f'(x)=\frac{1}{3}x^{-2/3}\Rightarrow f'(8)=\frac{1}{12}$, $f''(x)=-\frac{2}{9}x^{-5/3}\Rightarrow f''(8)=-\frac{2}{9\cdot 32}=-\frac{1}{144}$.

$$P_2(x)=2+\frac{x-8}{12}-\frac{(x-8)^2}{288},\qquad \sqrt[3]{8.3}\approx 2+0.025-0.0003125=2.0246875.$$

The calculator gives 2.024694.

7. a) Put $u=-x^2$: $e^{-x^2}\approx 1+(-x^2)+\frac{(-x^2)^2}{2}=1-x^2+\frac{x^4}{2}$.
b) $f'(x)=-2xe^{-x^2}$, $f''(x)=(4x^2-2)e^{-x^2}$, so $f''(0)=-2$. In the polynomial the coefficient of x^2 is $-1=\frac{f''(0)}{2!}$, so $f''(0)=-2$. It matches.
c) No. One term alone does not decide anything. What matters for raising the price is the sign of the derivative of the whole approximation over the relevant price range, plus the range where the approximation is valid and the economic constraints.

Problem answers · Topic 2.5

1. $\frac{6}{\sqrt{q}}=6q^{-1/2}$, so $C(q)=\int 6q^{-1/2}dq=\frac{6q^{1/2}}{1/2}+K=12\sqrt{q}+K$.
$C(100)=120+K=500\Rightarrow K=380$, so $C(q)=12\sqrt{q}+380$ and $C(400)=12\cdot 20+380=620$.

2. a) $\Delta C=\int_0^{50}20e^{0.02x}dx=\left[1000e^{0.02x}\right]_0^{50}=1000(e-1)\approx 1718.28$.
b) $C(x)=2000+\int_0^x 20e^{0.02t}dt=2000+1000\left(e^{0.02x}-1\right)=1000e^{0.02x}+1000$. Check: $C(0)=2000$.

3. $\int_0^3(x^2+ax+2)\,dx=\left[\frac{x^3}{3}+\frac{ax^2}{2}+2x\right]_0^3=9+\frac{9a}{2}+6$. Setting it equal to 18: $\frac{9a}{2}=3\Rightarrow a=\frac{2}{3}$.

4. Pieces of signed area: $[-2,0]$ triangle $+2$; $[0,2]$ rectangle $+4$; $[2,3]$ triangle $+1$; $[3,4]$ triangle -1; $[4,5]$ rectangle -2.
a) $\int_{-2}^5 f=2+4+1-1-2=4$ and $\int_0^4 f=4+1-1=4$.
b) Area $=2+4+1+1+2=10$. The integral counts the parts below the x-axis as negative; the area counts them as positive.

5. a) $e^x=e$ at $x=1$, and on $[0,1]$ the line $y=e$ is on top: $\int_0^1(e-e^x)\,dx=\left[ex-e^x\right]_0^1=(e-e)-(0-1)=1$.
b) $x^2=4$ at $x=\pm 2$: $\int_{-2}^2(4-x^2)\,dx=\left[4x-\frac{x^3}{3}\right]_{-2}^2=\frac{16}{3}+\frac{16}{3}=\frac{32}{3}$.

6. a) $|x-1|=1-x$ on $[-1,1]$ and $x-1$ on $[1,3]$: $\int_{-1}^1(1-x)dx+\int_1^3(x-1)dx=2+2=4$.
b) $\left\lfloor\frac{x}{2}\right\rfloor=0$ on $[0,2)$ and 1 on $[2,4)$: $0\cdot 2+1\cdot 2=2$.

7. $4f(x)-2g(x)=4F'(x)-2\big(2F'(x)+3x^2\big)=-6x^2$, so $H(x)=\int -6x^2\,dx=-2x^3+C,\ C\in\mathbb{R}$.

8. a) $f\ge 0$ needs $k\ge 0$, and $\int_0^3 kx^2dx=9k=1\Rightarrow k=\frac{1}{9}$.
b) $\int_0^1\frac{x^2}{9}dx=\frac{1}{27}\approx 0.037$.
c) $\frac{1}{3-0}\int_0^3 f(x)\,dx=\frac{1}{3}\cdot 1=\frac{1}{3}$.

9. a) $R(x) = 100x - 0.2x^2$ (with $R(0) = 0$). $R'(x) = 0 \iff x = 250$: maximum revenue $R(250) = 12\,500$.
b) $C(x) = 1000 + 20x + 0.1x^2$, $\pi(x) = R(x) - C(x) = 80x - 0.3x^2 - 1000$.
c) $\pi'(x) = 80 - 0.6x = 0 \iff x = \dfrac{400}{3} \approx 133.3$ units ($\pi'' = -0.6 < 0$). Maximum profit $\pi\left(\dfrac{400}{3}\right) = \dfrac{13\,000}{3} \approx 4333.33$.

10.

$$\frac{1}{10}\int_0^{10} \left(50 + 20e^{-0.1t}\right) dt = \frac{1}{10}\Big[50t - 200e^{-0.1t}\Big]_0^{10} = \frac{1}{10}\left(500 - 200e^{-1} + 200\right) = 70 - \frac{20}{e} \approx 62.64 \text{ units per day.}$$

Answers · past-paper style exam C

1. a) Continuity at 0: left $e^0 + a = 1 + a$, right $f(0) = 2$, so $a = 1$. Differentiability at 0: left slope $e^x \to 1$, right slope $2x + b \to b$, so $b = 1$. Continuity at 2: left $f(2) = 4 + 2 + 2 = 8$, right $\ln 1 + c = c$, so $c = 8$.
b) At $x = 2$: left slope $2x + 1 \to 5$, right slope $\dfrac{1}{x-1} \to 1$. Different: not differentiable at $x = 2$.
c) $\lim\limits_{x\to-\infty}(e^x + 1) = 1$: horizontal asymptote $y = 1$ on the left. $\lim\limits_{x\to+\infty}\big(\ln(x-1) + 8\big) = +\infty$: no asymptote on the right. No vertical asymptotes, since f is continuous on $\mathbb{R}$.

2. a) $\pi(q) = 100q - C(q) = -\dfrac{q^3}{3} + 12q^2 - 80q - 300$.
b) $\pi'(q) = -q^2 + 24q - 80 = -(q-4)(q-20) = 0 \iff q = 4$ or $q = 20$. $\pi''(q) = -2q + 24$: $\pi''(4) = 16 > 0$, local min; $\pi''(20) = -16 < 0$, local max.
c) $\pi(0) = -300$, $\pi(4) \approx -449.33$, $\pi(20) \approx 233.33$, $\pi(30) = -900$. Global maximum €233.33 at $q = 20$; global minimum €−900 at $q = 30$ (producing at full capacity is the worst option).
d) $\pi''(q) = 0 \iff q = 12$, with a sign change: inflection point at $q = 12$. Marginal profit π' is increasing before 12 units and decreasing after, so $q = 12$ is where marginal profit is highest.

3. a) $1 + 2x > 0 \iff x > -\dfrac{1}{2}$: $D = \left(-\dfrac{1}{2}, +\infty\right)$.
$f(0) = 0$, $f'(x) = \dfrac{2}{1+2x} \to 2$, $f''(x) = -\dfrac{4}{(1+2x)^2} \to -4$, $f'''(x) = \dfrac{16}{(1+2x)^3} \to 16$:

$$P_3(x) = 2x - 2x^2 + \frac{8}{3}x^3.$$

b) $x = 0.1$: $P_3(0.1) = 0.2 - 0.02 + 0.002667 = 0.182667$, while $\ln 1.2 \approx 0.182322$.
c) Tangent $y = 2x$. The first term after it, $-2x^2$, is negative: f lies below its tangent on both sides, concave downwards near 0.

4. a) $R(q) = 200q - 0.4q^2$, $C(q) = 3000 + 40q + 0.1q^2$.
b) $\pi'(q) = R'(q) - C'(q) = 160 - q = 0 \iff q = 160$ (and $\pi'' = -1 < 0$).
c) $\pi(160) = \displaystyle\int_0^{160} (160 - q)\, dq - 3000 = \Big[160q - \frac{q^2}{2}\Big]_0^{160} - 3000 = 12\,800 - 3000 = 9800$.
d) $\displaystyle\int_{100}^{160} (200 - 0.8q)\, dq = \Big[200q - 0.4q^2\Big]_{100}^{160} = 21\,760 - 16\,000 = 5760$.

Answers · past-paper style exam D

1. a) Mathematical domain $\mathbb{R}\setminus\{120\}$; contextual domain $[0, 100]$ (a percentage). Since 120 is outside $[0, 100]$, the cost stays finite: removing everything costs $C(100) = \dfrac{800\,000}{20} = 40\,000$.

b) $C'(x) = \dfrac{8000(120-x)+8000x}{(120-x)^2} = \dfrac{960\,000}{(120-x)^2} > 0$: increasing.

c) $C'(60) = \dfrac{960\,000}{3600} \approx 266.67$ per percentage point: at 60% removed, one more point costs about €266.67.

d) $C(60) = 8000$. $C(61) \approx 8000 + 266.67 = 8266.67$. Exact: $C(61) = \dfrac{488\,000}{59} \approx 8271.19$; the estimate is €4.52 too low.

2. a) $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$. Increasing on $(-2,-1)$ and $(1,3)$, decreasing on $(-1,1)$. Local max $f(-1) = 2$, local min $f(1) = -2$.

b) $f''(x) = 6x = 0 \iff x = 0$, sign change: inflection point $(0,0)$.

c) $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$. Global maximum 18 at $x = 3$ (an endpoint); global minimum -2 at $x = -2$ and $x = 1$.

3. a) $f(-1) = 5$, $f'(-1) = 0$, $\dfrac{f''(-1)}{2} = -3 \Rightarrow f''(-1) = -6$, $f'''(-1) = 0$, $\dfrac{f^{(4)}(-1)}{24} = 2 \Rightarrow f^{(4)}(-1) = 48$. Tangent: $y = 5$ (horizontal). The first term after it is $-3(x+1)^2 \le 0$: f is below its tangent, so $x = -1$ is a **local maximum** (concave down).

b) $\dfrac{h'''(2)}{6} = -\dfrac{1}{2} \Rightarrow h'''(2) = -3$. Tangent: $y = -1 + 2(x-2)$. $P_3 - P_1 = -\dfrac{1}{2}(x-2)^3$ is an odd power: positive for $x < 2$ (h above its tangent) and negative for $x > 2$ (below). Inflection point at $x = 2$.

4. a) $\displaystyle\int_{-2}^{2} k(4-x^2)\,dx = k\Big[4x - \frac{x^3}{3}\Big]_{-2}^{2} = \frac{32k}{3} = 1 \Rightarrow k = \frac{3}{32}$ (and $f \ge 0$).

$\displaystyle\int_{-1}^{1} \frac{3}{32}(4-x^2)\,dx = \frac{3}{32}\cdot\frac{22}{3} = \frac{11}{16} = 0.6875.$

b) $x^2 = x + 2 \iff x = -1$ or $x = 2$; on $[-1,2]$ the line is on top:

$$\int_{-1}^{2} (x + 2 - x^2)\,dx = \Big[\frac{x^2}{2} + 2x - \frac{x^3}{3}\Big]_{-1}^{2} = \frac{10}{3} + \frac{7}{6} = \frac{9}{2}.$$